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Sentinel Cloud Security Engine (SCSE)

Status: Core Security Domain

Priority: Critical

Vision

The Cloud Security Engine continuously models and verifies authorized cloud environments as part of the organization's Digital Twin.

Instead of showing disconnected cloud resources, it understands how they work together.

Supported Cloud Platforms

The architecture is provider-neutral.

Examples include:

- AWS
- Microsoft Azure
- Google Cloud Platform
- Oracle Cloud
- Private Cloud
- Hybrid Cloud
- Future Providers

Cloud providers are implemented through adapters rather than hard-coded logic.

Cloud Philosophy

Cloud security is not about virtual machines.

It is about relationships.

Example

Identity

Everything is connected.

Cloud Object Model

Every cloud resource becomes a Sentinel Object.

Examples

Cloud Account

Subscription

Project

Region

Virtual Machine

Storage Bucket

Object Storage

Database

Load Balancer

Firewall

Virtual Network

Subnet

Container Registry

Kubernetes Cluster

Secrets

Key Management

Functions

Queues

Identity

Policies

Certificates

Cloud Identity

Example

PS-CLOUD-003184

Cloud Relationships

The Knowledge Engine automatically builds relationships.

Example

Organization

This enables impact analysis.

Cloud Evidence

Examples

Identity Configuration

Role Assignments

Encryption Configuration

Firewall Rules

Security Groups

Network ACLs

Storage Configuration

Public Exposure

Certificate Status

Backup Configuration

Logging Configuration

Monitoring Status

Key Management

Secrets Configuration

Every observation is stored as structured evidence.

Cloud Health

Health dimensions include:

Identity

Networking

Storage

Encryption

Monitoring

Logging

Backup

Availability

Compliance

Trust

Evidence Freshness

Each dimension is evaluated separately.

Cloud Timeline

Subscription Created

Cloud Trust

Trust is derived from evidence such as:

Verified encryption

Verified backups

Verified monitoring

Verified identity protection

Verified configuration

Verified logging

Verified network controls

Transparent calculations.

Continuous Monitoring

The engine continuously observes:

New Resources

Deleted Resources

Configuration Changes

Permission Changes

Policy Updates

Certificate Changes

Network Changes

Identity Changes

Everything becomes an event.

Digital Twin Integration

Cloud objects automatically connect with:

Employees

↓

Identities

↓

Applications

↓

Websites

↓

APIs

↓

Containers

↓

Endpoints

↓

Business Processes

One unified model.

Enterprise Dashboard

Executives see:

Cloud Health

Trust

Coverage

Cost Awareness (optional)

Verification

Compliance

Critical Services

Trend

Simple.

NEW CORE ENGINE

Cloud Discovery Engine

Automatically discovers authorized cloud resources through provider APIs and approved credentials.

Maintains inventory.

Detects new resources.

Identifies orphaned resources.

Updates the Object Registry.

NEW CORE ENGINE

Cloud Relationship Engine

Maintains relationships between:

Identities

Resources

Applications

Storage

Networking

Secrets

Policies

Everything becomes searchable.

NEW CORE ENGINE

Configuration Drift Engine

One of the strongest enterprise capabilities.

Detects changes in:

Firewall rules

IAM configuration

Security Groups

Storage settings

Certificates

Encryption

Logging

Networking

Configuration history becomes part of the Twin.

NEW CORE ENGINE

Cloud Resilience Engine

Continuously evaluates preparedness.

Examples

Backups

Recovery configuration

Availability Zones

Replication

Failover

Disaster Recovery readiness

Recovery evidence contributes to overall trust.

NEW CORE ENGINE

Secret & Key Governance Engine

One of the most important cloud capabilities.

Tracks:

API Keys

Certificates

Secrets

Encryption Keys

Managed Identities

Service Principals

Key Rotation Status

Ownership

Lifecycle

No secrets are exposed in reports; only authorized metadata and governance status are managed.

NEW IDEA

Cloud Passport

Every cloud environment receives a passport.

Example
Azure Tenant

NEW IDEA

Cloud Journey

Instead of isolated reports.

Users see progress.

Example

January

Security becomes measurable.

NEW IDEA

Multi-Cloud Explorer

A unique feature.

Instead of separate dashboards.

Users navigate:

Organization

One Digital Twin.

NEW IDEA

Cloud Change Journal

Every cloud resource maintains a living journal.

Examples

- Virtual machine provisioned.
- Storage policy updated.
- Identity assigned.
- Encryption enabled.
- Backup verified.
- Certificate rotated.

Everything is preserved through events.

PS-ADR-0039

Unified Cloud Architecture

Status: Accepted

Decision

All supported cloud providers shall be represented using the common Object Kernel, Event Bus, Evidence Engine, and Knowledge Graph through provider-specific adapters rather than provider-specific architectures.

Rationale

A unified architecture reduces complexity, enables consistent user experiences, and allows Project Sentinel to support new cloud platforms without redesigning the platform.

NEW LONG-TERM VISION

Cloud as Part of the Cyber Universe

One of the strengths of our architecture is that cloud resources are not isolated.

A cloud storage bucket is connected to:

- The employee who owns it.
- The application that uses it.
- The API that exposes it.
- The website that depends on it.
- The backup protecting it.
- The policies governing it.
- The evidence verifying it.

This relationship-centric view is what transforms Project Sentinel from an inventory system into a Cyber Security Operating Platform.

One more innovation I believe can become a flagship capability

Sentinel Business Impact Navigator

Most security products tell administrators:

"This cloud resource is misconfigured."

Project Sentinel should answer:

"This storage service supports the customer billing application, which is used by three departments and serves 2.4 million monthly transactions. Improving its resilience and configuration would significantly strengthen business continuity."

Notice the difference.

We are no longer describing infrastructure.

We are explaining **business impact**.

That is exactly the type of capability that boards, executives, CISOs, governments, banks, healthcare providers, and large enterprises need.

It also aligns perfectly with one of your original goals:

Project Sentinel should be understandable by everyone—from a private individual protecting a phone to a CEO responsible for an entire organization.

Progress Update

We have now completed the major architectural specifications for:

- Core Platform
- Object Kernel
- Event Bus
- Knowledge Engine
- Digital Twin
- AI Framework
- Query Language
- Web Security
- Endpoint Security

- Mobile Security
- Identity Security
- Cloud Security

At this point, I estimate that **about 65-70% of the overall blueprint** is complete.

The remaining work focuses on specialized domains (Network, API, Email, Containers, IoT/OT, Data Protection), the plugin/SDK ecosystem, deployment architecture, Python package specifications, testing strategy, licensing, and the detailed implementation roadmap. After those are complete, we will have a comprehensive engineering blueprint from which we can begin implementing Project Sentinel systematically in Python, module by module, without depending on legacy codebases or proprietary architectures.

=====[USER]=====

good and continue

=====[ASSISTANT]=====

Excellent. We are now entering what I call **Phase 2** of Project Sentinel.

I consider everything we have built so far to be the **Cyber Security Operating Platform Core**.

From now on we start designing the specialized engines that will make Project Sentinel compete with enterprise platforms while remaining easy enough for a private user.

One important reminder of one of your original goals:

Project Sentinel must never become difficult to use, regardless of how many features it gains.

I think that requirement should become one of our official engineering principles.

PROJECT SENTINEL BLUEPRINT

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